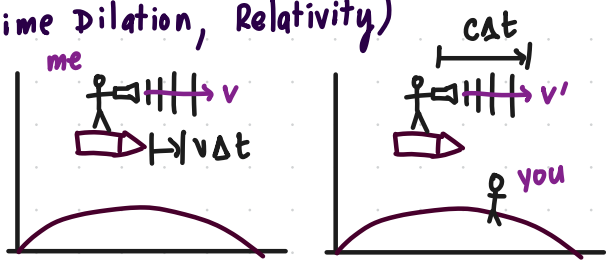
Lorentz Transformation (Time Dilation, Relativity)

my frame

$$\Delta x = \gamma(\Delta x + v\Delta t')$$

$$\Delta t = \gamma\left(\Delta t' + \frac{v}{c^2}\Delta x'\right)$$



your frame

$$\Delta x' = \gamma(\Delta x - v\Delta t)$$

$$\Delta t' = \gamma\left(\Delta t - \frac{v}{c^2}\Delta x\right)$$

$$\left.\begin{aligned}\Delta x' &= \gamma(c\Delta t - v\Delta t)\\ \Delta t' &= \gamma\left(\Delta t - \frac{v}{c^2}c\Delta t\right)\end{aligned}\right\}$$

speed of light seen by you:

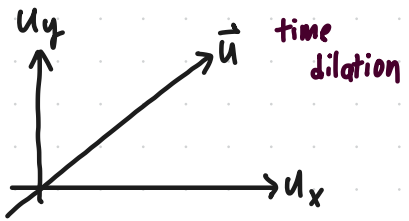
$$u' = \frac{\Delta x'}{\Delta t'} = \frac{\cancel{\gamma}(c\cancel{\Delta t} - v\cancel{\Delta t})}{\cancel{\gamma}\left(\cancel{\Delta t} - \frac{v}{c^2}c\cancel{\Delta t}\right)} = \frac{c}{c}\frac{(c-v)}{\left(1-\frac{v}{c}\right)}$$

$$= \frac{c(c-v)}{c-v} = \underline{c}$$

velocity transformation

$$u = \frac{\Delta x}{\Delta t} = \frac{\gamma(\Delta x' + v\Delta t')}{\gamma\left(\Delta t' + \frac{v}{c^2}\Delta x'\right)}\frac{\left(\frac{1}{\Delta t'}\right)}{\left(\frac{1}{\Delta t'}\right)}$$

$$*u' = c \qquad \hookrightarrow = \frac{\left(\frac{\Delta x'}{\Delta t'} + v\right)}{\left(1 + \frac{v}{c^2}\frac{\Delta x'}{\Delta t'}\right)} \Rightarrow \frac{u_x' + c}{1 + \frac{vu_x'}{c^2}} = \frac{c+v}{1+\frac{v\cancel{c}}{\cancel{c^2}}}\frac{c}{c} = \underline{c}$$



time dilation

length contraction

length contraction:

$$u_y = \frac{\Delta y}{\Delta t} = \frac{\Delta y'}{\gamma\left(\Delta t' + \frac{v}{c^2}\Delta x'\right)}\frac{\left(\frac{1}{\Delta t'}\right)}{\left(\frac{1}{\Delta t'}\right)} \Rightarrow \frac{\frac{\Delta y'}{\Delta t'}}{\gamma\left(1 + \frac{v}{c^2}\frac{\Delta x'}{\Delta t'}\right)} = \frac{\Delta y'}{\gamma\left(1 + \frac{vu_y}{c^2}\right)}$$

example of an invariant:

uni

$$\boxed{(\Delta s^2)^2 = (c\Delta t)^2 - (\Delta x)^2} = c^2\gamma^2\left(\Delta t' + \frac{v}{c^2}\Delta x'\right)^2 - \gamma^2(\Delta x' + v\Delta t')^2$$

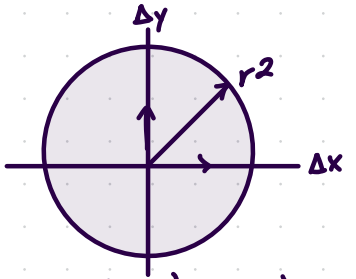
$$= c^2\gamma^2\left[(\Delta t')^2 + \frac{2v}{c^2}\Delta x'\Delta t' + \frac{v^2}{c^4}(\Delta x')^2\right] - c^2\gamma^2\left[\frac{(\Delta x')^2}{c^2} + 2\frac{v}{c^2}\Delta x'\Delta t' + \frac{v^2}{c^2}(\Delta t')^2\right]$$

$$\Rightarrow c^2\cancel{\gamma^2}\left[(\Delta t')^2 - \cancel{\frac{v^2}{c^2}}(\Delta t')^2\right] + c^2\gamma^2\left[\frac{v^2}{c^4}(\Delta x')^2 - (\Delta x')^2\right]$$

$$\hookrightarrow = \frac{c^2\gamma^2}{c^2}\left(\frac{v^2}{c^2} - 1\right)(\Delta x)^2$$

$$(\Delta s^2)^2 = (c\Delta t)^2 - (\Delta x)^2 = \frac{c^2\gamma^2}{c^2}\left(\frac{v^2}{c^2} - 1\right)(\Delta x)^2$$

"space-time"

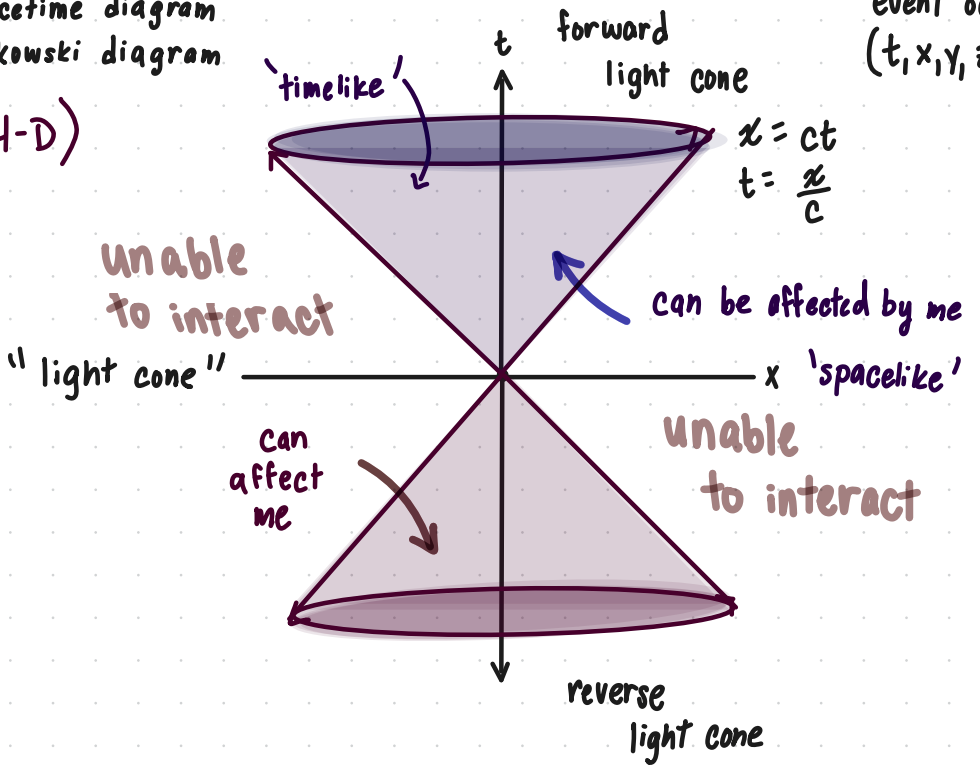


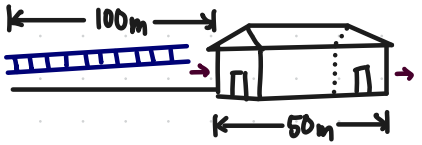
$r^2 = (\Delta x^2) + (\Delta y^2)$

spacetime diagram
minkowski diagram

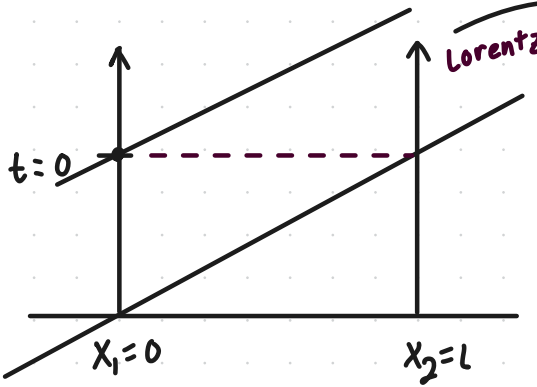
(4-D)

event occurs (t, x, y, z) $\xrightarrow{\text{Lorentz transform}}$ event from other reference frame (t', x', y', z')

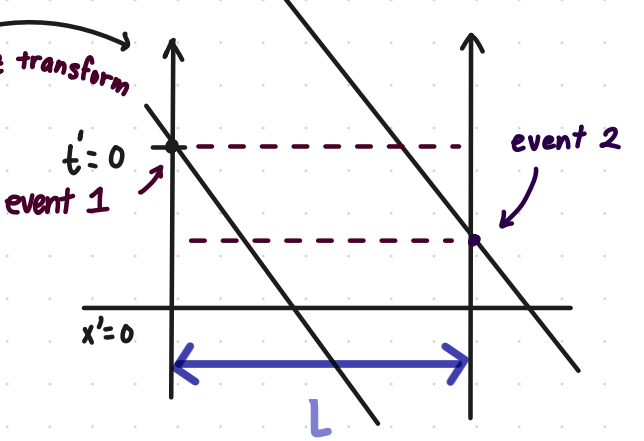




(L)



Lorentz transform



from one reference:
time dilation

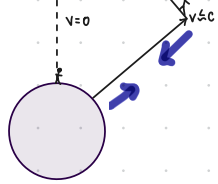
from other:
length contraction

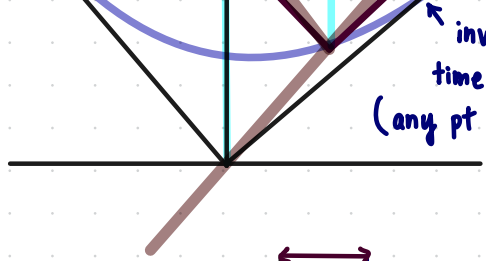
$$t_1' = \gamma\left(0 - \frac{v}{c^2}(0)\right) = 0$$

$$x_1' = \gamma(0 - 0) = 0$$

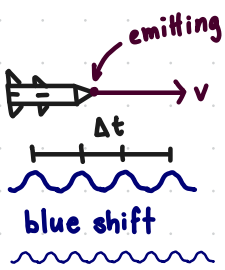
$$t_2' = \gamma\left(0 - \frac{v}{c^2}(L)\right) = -\frac{v}{c^2}L^2$$

$$x_2' = \gamma(L - v0) = \gamma L$$





invariant time curve
(any pt on this curve experiences same universal time)



in frame of emitter

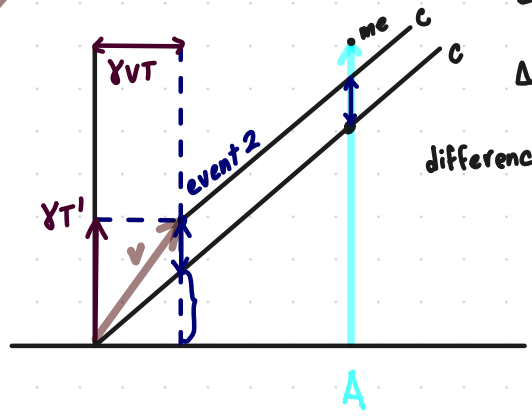
prime frame
$$\begin{cases} \text{emit 1: } x' = 0,\ t' = 0 \\ \text{emit 2: } x' = 0,\ t' = T \end{cases}$$

our frame

emit 1: $x = 0$, $t = 0$

emit 2: $x_2 = \gamma(0 + vT)$

$t_2 = \gamma(T + 0)$



$\Delta t_1 = \frac{\gamma v T'}{c}$ (time for 1st pulse)

$\Delta t_2 = \gamma T'$ (emission of 2nd pulse)

difference: $T = \gamma T' - \frac{\gamma v T'}{c}$

$= \gamma\left(1 - \frac{v}{c}\right)T'$

$\Rightarrow \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}\left(1 - \frac{v}{c}\right)T' \Rightarrow \frac{\left(1 - \frac{v}{c}\right)}{\sqrt{\left(1 - \frac{v}{c}\right)\left(1 + \frac{v}{c}\right)}}T'$

$\hookrightarrow \sqrt{\frac{1 - \frac{v}{c}}{1 + \frac{v}{c}}}\,T'$

frequency: $\frac{1}{T} = \frac{1}{T'}\sqrt{\frac{1 + v/c}{1 - v/c}}$

invariant:

$$E^2 - (pc)^2 \Rightarrow E = \gamma mc^2$$

$$p = \gamma mv$$

kinetic energy

momentum

$\llcorner\ mv$

clarification! $-\frac{1}{2}mv^2$

$$E = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}\, mc^2$$

Taylor: $f(x_0 + x) = f(x_0) + \left.\frac{\partial f}{\partial x}\right|_{x_0} x + \dots$

$$\hookrightarrow \frac{1}{\sqrt{1-x}} \Rightarrow 1 - \frac{1}{2}(1-x)^{-3/2}(-1) + \dots$$

$(x_0 = 1)$

$$= 1 + \frac{1}{2}x$$

$$\frac{1}{\sqrt{1-\frac{v^2}{c^2}}} \cong \left(1 + \frac{1}{2}\frac{v^2}{c^2}\right)mc^2$$

$$\cong \underline{mc^2} + \boxed{\frac{1}{2}mv^2}$$

"rest energy"